

CATHOLIC JUNIOR COLLEGE
JC2 Preliminary Examinations
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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BIOLOGY

9648/03

Applications Paper and Planning Question

25 August 2015

Paper 3

2 hours

Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Section A	
1	
2	
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4	
5	
Total	

This document consists of **14** printed pages and **2** blank pages.

[Turn over

Answer **all** questions.

- 1 The artificial plasmid, pBR322, was constructed to act as a vector. It has often been used to insert human genes, such as the human insulin gene, into the bacterium, *Escherichia coli*.

The plasmid was constructed to include two genes, each giving resistance to a different antibiotic: an ampicillin resistance gene and a tetracycline resistance gene. The plasmid also has a target site for the restriction enzyme, *Bam*HI, in the middle of the tetracycline resistance gene.

A pBR322 plasmid was cut using *Bam*HI and the cDNA gene for human insulin inserted into it. Fig. 1.1 shows pBR322 and the recombinant plasmid.

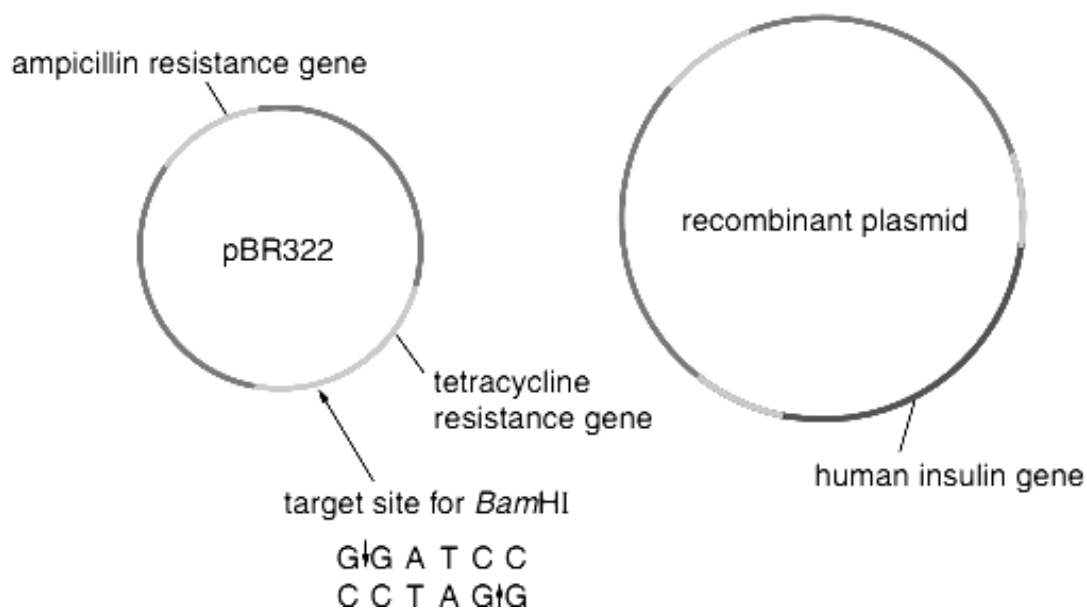


Fig. 1.1

- (a) With reference to Fig. 1.1, describe how a cDNA human insulin gene can be inserted into pBR322 that has been cut by *Bam*HI.

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.....[4]

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[Turn over

(b) Bacteria were then mixed with the recombinant plasmids. Those bacteria which had successfully taken up recombinant plasmids were identified using the following steps:

- step 1 – the bacteria were spread onto culture plates containing nutrient agar and ampicillin and incubated to allow colonies to form
- step 2 – some bacteria from each of the colonies growing on these plates were transferred to plates containing nutrient agar and tetracycline, as shown in Fig. 1.2.

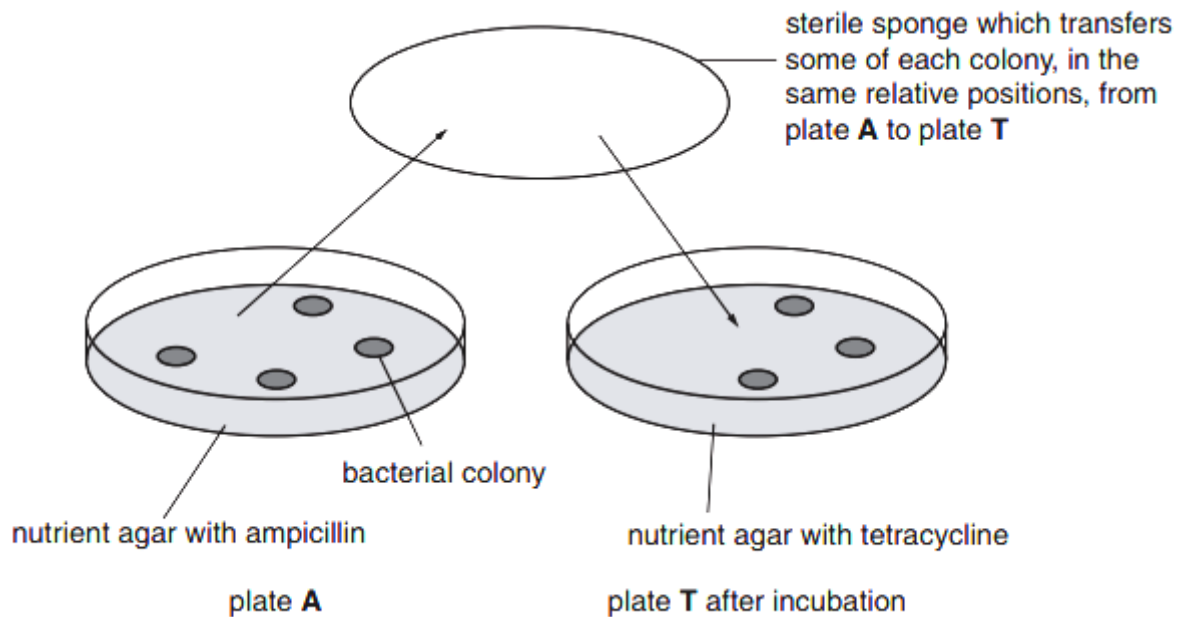


Fig. 1.2

(i) Explain why the bacteria were first spread onto plates containing ampicillin.

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.....[3]

- (ii) Explain why it is important, for identifying bacteria that have successfully taken up the recombinant plasmid, that on pBR322 the target site for *Bam*HI is in the middle of the tetracycline resistance gene.

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.....[3]

- (iii) Use a label line and the letter **C** to identify, on Fig. 1.2, a colony of bacteria that contain the recombinant plasmid.

Put your answer onto Fig. 1.2 on page 4.

[1]

- (c) Plasmid vectors carrying antibiotic resistance genes are now rarely used in gene technology.

- (i) Suggest why antibiotic resistance genes are now rarely used.

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.....[2]

- (ii) Suggest one type of gene that has replaced antibiotic resistance genes in plasmid vectors and indicate how its presence can be detected.

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[Total: 15]

[Turn over

2 Stem cells: recent progress in medical application.

In 2008, a woman had her damaged left bronchus replaced by one constructed using her own stem cells. A section of the trachea from a donor was treated in order to remove the cells of the donor, leaving only a framework of collagen. Adult stem cells derived from the woman's own bone marrow were then added to this collagen framework. A month later the transplanted tissue had developed its own blood supply. This was claimed to be the first successful transplant using tissues derived from the patient's own stem cells.

- (a) Outline the advantages of using a patient's own stem cells to treat a damaged bronchus, as described in the passage, rather than transplanting a complete bronchus.

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.....[3]

- (b) Adult stem cells derived from the patient's own bone marrow are described as multipotent, rather than totipotent.

Distinguish between multipotent cells and totipotent cells.

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.....[1]

- (c) Once most stem cells differentiate, they lose their ability to proliferate indefinitely and differentiate into other types of cells. However, some fully differentiated cells can be stimulated to change back into stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells). iPS cells regain the unique properties of stem cells.

In experiments with mice it was discovered that the introduction of four genes would cause certain fully differentiated cells to change to iPS cells.

There is evidence to suggest that the introduction of the four genes has caused an increase in the production of telomerase reverse transcriptase (TERT) in the fully differentiated cells.

- (i) State how the addition of the four genes caused an increase in the production of TERT.

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.....[1]

- (ii) Explain how TERT may help to change the cells back into stem cells.

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.....[2]

[Turn over

A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 2.1.

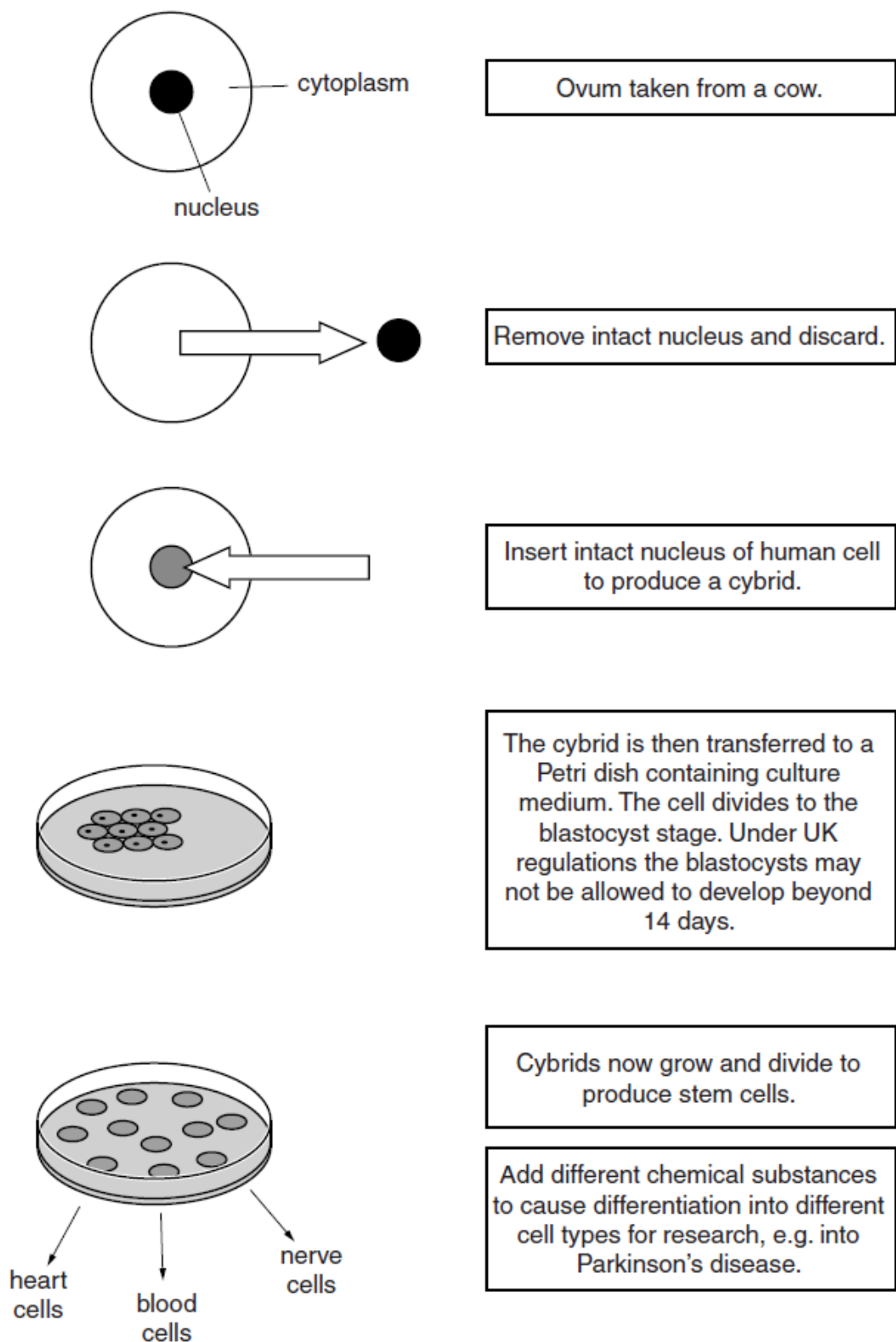


Fig. 2.1

- (d) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is in the cytoplasm. Explain the presence of the 0.4% DNA in the cytoplasm.

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.....[2]

- (e) Some people argued that it is unethical to allow the production of cybrids.

State whether you agree **or** disagree that this is unethical **and** explain why you reached this decision.

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[Total: 11]

[Turn over

- 3 (a) To produce *Bt* maize, the procedure to genetically modify plant cells, shown in Fig.3.1, was carried out.

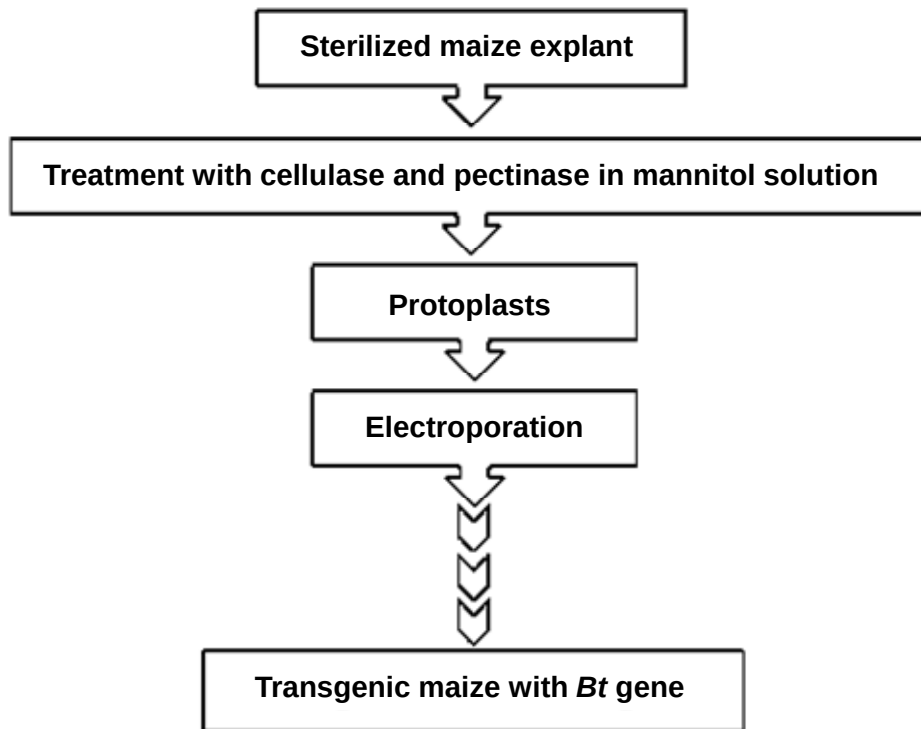


Fig. 3.1

- (i) Name a vector suitable for genetically modifying plant cells that can be used in this procedure.

.....[1]

- (ii) Explain why the sterilized explant was treated with cellulase and pectinase before electroporation.

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[2]

Addition of mannitol to water lowers the water potential of water. In the preparation of protoplasts, mannitol must be present. In the absence of mannitol, no protoplasts are formed even if the sterilised explant was treated with cellulase and pectinase.

- (iii) State what happens to the protoplasts if mannitol was not used.

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[1]

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Fig. 3.2 shows the proportion of cotton, maize and soybeans grown in the USA that are genetically modified in two different ways:

- HT crops are modified to be resistant to the herbicide glyphosate.
- *Bt* crops are modified to express the *Bt* toxin which kills insect pests.

percentage of total
crop grown that is
genetically modified

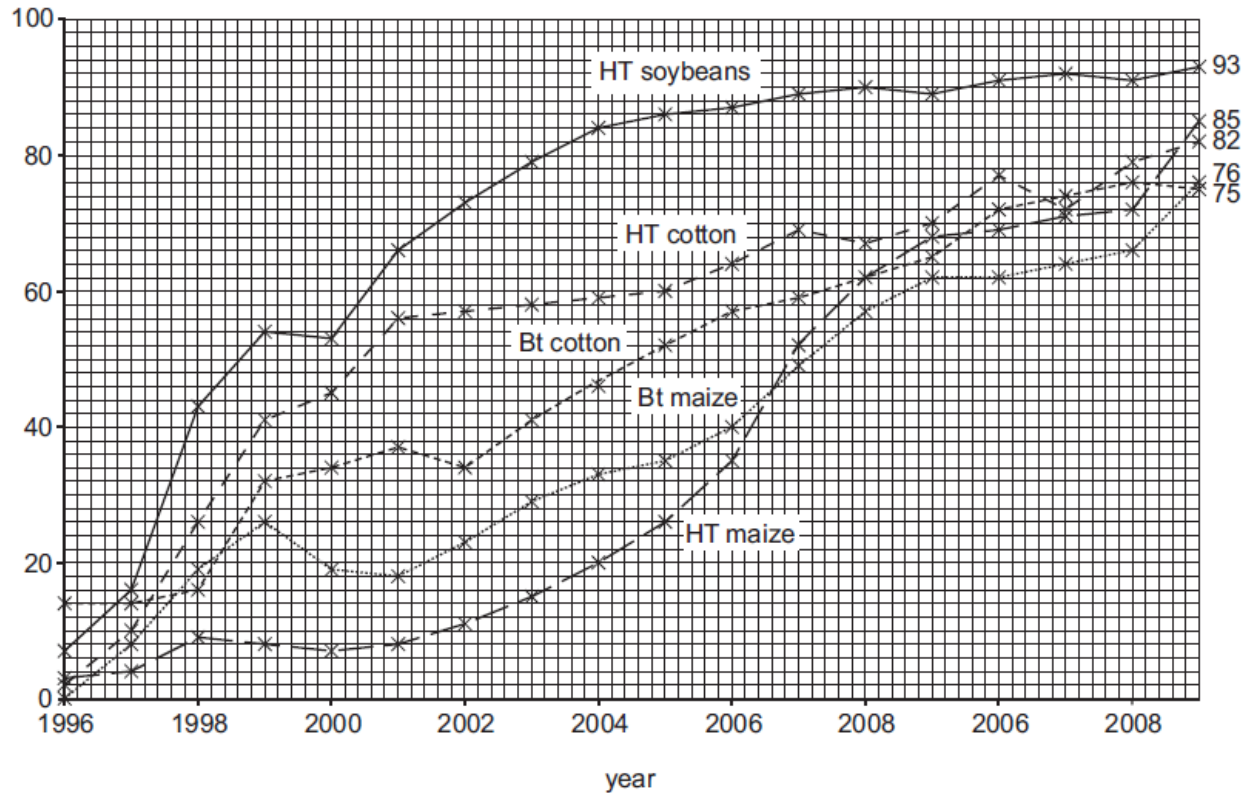


Fig 3.2

(b) With reference to Fig. 3.2, describe the change in the percentage of genetically modified cotton grown in the USA since 1996.

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.....[2]

- (c) The HT crops received two new genes that gave resistance to glyphosate herbicide and also a marker gene called GUS. The parts of the plant that express the GUS gene turn blue when dipped into a colourless chemical substrate.

Explain why the GUS gene was also transferred to the genetically modified crops.

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Turn over]

- (d) Fig. 3.3 shows the increase in the number of weed species resistant to glyphosate herbicide and triazine herbicides since 1970. Crops have not been genetically modified to resist triazine herbicides.

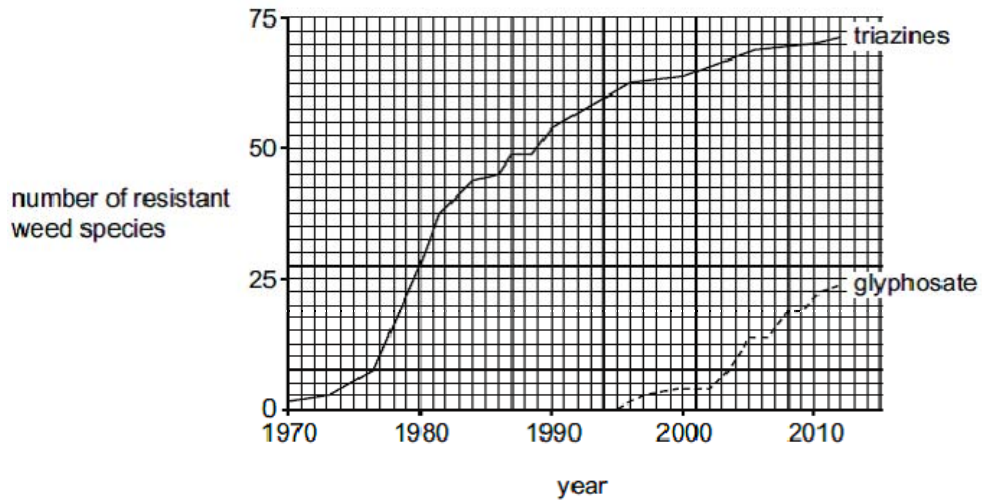


Fig 3.3

- (i) With reference to Fig. 3.2 and Fig. 3.3, justify the statement that the rise in glyphosate-resistant weeds has resulted from the introduction of genetically modified crops.

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- (ii) Give **one** piece of evidence from Fig. 3.3 to support the idea that the development of weeds resistant to herbicides is **not** due to horizontal gene transfer from genetically modified crops.

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- (iii) Outline how herbicide resistance in weeds arises and spreads without horizontal gene transfer.

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[Total: 14]

4 Planning question

The enzyme urease is a catalyst for the hydrolysis of urea in solution, for example in the breakdown of urea in soils by microorganisms, forming ammonia and carbon dioxide.

As the reaction proceeds, the ammonia released dissolves, causing the pH to increase.

You are required to plan an investigation to compare the activity of urease free in solution and urease immobilised in alginate beads.

You are provided with the following equipment which you may or may not use in your plan. However, you may **not** use any additional equipment in your plan.

- calcium alginate beads all of uniform size prepared with a 50gdm^{-3} urease solution (you may call this immobilised urease)
- 50gdm^{-3} urease solution (you may call this free urease)
- 1.0mol dm^{-3} urea solution
- distilled water
- beakers and flasks of different sizes
- stop watch or electronic timer
- pH probes and meters
- thermometer
- thermostatically-controlled water baths
- graduated pipettes and pipette fillers
- filter funnels
- syringes
- glass rods for stirring
- test-tubes and boiling tubes
- test-tube racks

Your plan should

- include a clear statement of the hypothesis or prediction
- identify the key variables
- give full details and explanations of the procedures that you would adopt to ensure that the results are as precise and reliable as possible
- show how you would present and analyse your results
- include a brief risk assessment
- be written in clear scientific language.

You may include a diagram or diagrams in your plan.

[Total: 12]

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Explain the different uses of RFLP analysis. [6]

(b) Describe the advantages and limitations of plant tissue culture as compared to traditional agricultural methods such as grafting and cutting. [8]

(c) Discuss the benefits and ethical concerns of the human genome project. [6]

[Total: 20]